Huawei Access Node HEB-ACC-42 QoS Congestion Event Alarm ALARM_7936 Resolution SOP

Node: HEB-ACC-42 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7936

Description

This Standard Operating Procedure (SOP) addresses a Warning-level QoS congestion event alarm (ALARM_7936) on Huawei access node HEB-ACC-42 at the Hebbal Edge site. QoS congestion events, often manifesting as microbursts, interface tail drops, or QoS queue starvation, occur due to the node's inability to process traffic at a rate commensurate with the offered load. This may be attributed to the node's hardware configuration, such as inadequate buffer sizes or misconfigured traffic policing profiles, or software-related issues like inadequate QoS policies or faulty routing protocols. The congestion event can lead to packet loss, increased latency, and degradation of overall network performance. The symptoms of this issue can be further exacerbated by the presence of misconfigured traffic shaping or policing mechanisms, which can lead to a decrease in the node's ability to handle the traffic, thus resulting in the aforementioned congestion event.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that remote hands are dispatched to the location.

- Check the status of redundant links to ensure they are operational and not contributing to the congestion event.

- Review the node's configuration to ensure that QoS policies are properly implemented and that traffic policing profiles are correctly set.

Potential Root Causes

Cause	Probability
Inadequate QoS policies or misconfigured traffic policing profiles	High
Faulty routing protocols or misconfigured traffic shaping mechanisms	Medium
Hardware limitations, such as inadequate buffer sizes	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policer-profile, Expected Output: Policer Profile Information, Description: This command displays the policer profile information, which can help identify if the policer profile is correctly configured. The command can isolate if the policer is correctly set to discard traffic that exceeds the configured rate.

Command: display interface all, Expected Output: Interface Information, Description: This command displays the interface information, which can help identify if the interface is experiencing congestion. The command can isolate if the interface is experiencing
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interface tail drops, which can be a symptom of QoS congestion events.

Command: display qos queue, Expected Output: QoS Queue Information, Description: This command displays the QoS queue information, which can help identify if the queue is experiencing starvation. The command can isolate if the queue is experiencing QoS queue starvation, which can be a symptom of QoS congestion events.
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Resolution Steps

Step 1: Step: Review and adjust QoS policies, Command: config qos policer-profile, Description: Review the QoS policies and adjust them as necessary to ensure that the policer profile is correctly configured to discard traffic that exceeds the configured rate.

Step 2: Step: Adjust traffic policing profiles, Command: config policer-profile, Description: Adjust the traffic policing profiles to ensure that they are correctly set to discard traffic that exceeds the configured rate.

Step 3: Step: Verify routing protocols and traffic shaping mechanisms, Command: display routing-table, Description: Verify that the routing protocols are correctly configured and that the traffic shaping mechanisms are correctly set to prevent congestion events.

Post-Check Verification

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Command: display qos policer-profile, Expected Output: Policer Profile Information, Description: Verify that the policer profile is correctly configured and that traffic is being policed correctly.

Command: display interface all, Expected Output: Interface Information, Description: Verify that the interface is not experiencing congestion and that interface tail drops are not occurring.

Command: display qos queue, Expected Output: QoS Queue Information, Description: Verify that the queue is not experiencing starvation and that QoS queue starvation is not occurring.
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Roll-back Procedures

- Step: Restore previous QoS policies, Command: undo config qos policer-profile, Description: Restore the previous QoS policies in case the fix fails.

- Step: Restore previous traffic policing profiles, Command: undo config policer-profile, Description: Restore the previous traffic policing profiles in case the fix fails.

- Step: Restore previous routing protocols and traffic shaping mechanisms, Command: undo config routing-table, Description: Restore the previous routing protocols and traffic shaping mechanisms in case the fix fails.